FIVE EQUIVALENT FORMULATIONS OF PROPERTY (P) OF A RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Benign-screen note: the proof contains the phrase 'it is clear that' / 'is obvious' once, used as a legitimate step inside a complete argument; the author proves the statement, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- Property (P) implies universal injectivity of injective maps of projective modules. Lemma \ref{lemma-P-universally-injective}, cited at proof lines 3773-3774. Gives the first implication.
- Universally injective maps let one split inclusions of finite projective modules. Algebra, Lemma \ref{algebra-lemma-universally-exact-split}, cited at proof lines 3777-3778. Gives the middle implications.
- The definition of property (P) through finitely generated ideals, and Nakayama's lemma. Lemma \ref{lemma-auto-ass-implies-P}, cited at proof line 3757, and Algebra, Lemma \ref{algebra-lemma-NAK}, used at proof lines 3779-3784. Close the cycle from the last formulation.

Source reference. Original lemma 05GQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7069–7069. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 05GN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7067–7067. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05GP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7068–7068. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 058L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6784–6784. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. The following are equivalent*

(1) *R has property (P) of Lemma 05GN,*
(2) *any injective map of projective R-modules is universally injective,*
(3) *if $u : N \to M$ is injective and N, M are finite projective R-modules then* $\mathrm{Coker}(u)$ *is a finite projective R-module,*
(4) *if $N \subset M$ and N, M are finite projective as R-modules, then N is a direct summand of M, and*
(5) *any injective map $R \to R^{\oplus n}$ is a split injection.*

Proof. The implication (1) $\Rightarrow$ (2) is Lemma 05GP. It is clear that (3) and (4) are equivalent. We have (2) $\Rightarrow$ (3), (4) by Algebra, Lemma 058L. Part (5) is a special case of (4). Assume (5). Let $I = (a_1, \ldots, a_n)$ be a proper finitely generated ideal of R. As $I \neq R$ we see that $R \to R^{\oplus n}$, $x \mapsto (xa_1, \ldots, xa_n)$ is not a split injection. Hence it has a nonzero kernel and we conclude that $\mathrm{Ann}_R(I) \neq 0$. Thus (1) holds. $\square$